

Anthony Mwangela Robert

Software Engineer · Full-Stack & DevOps · Founder, Robertech

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Portfolio robadev254-robertecch-vmgj.vercel.app | GitHub github.com/Robadev254 | LinkedIn in/anthony-robert-5a5290227

PROFESSIONAL SUMMARY

Software engineer with a BSc in Software Development from KCA University (Second Class Honours, Upper Division) and a distinction-grade ALX Software Engineering qualification. I work across the full stack — accessible responsive interfaces, REST APIs and relational data models, and the Linux systems and infrastructure-as-code underneath them. Most recently IT Support Personnel at Kenya Forest Service headquarters, covering database administration, network installation and troubleshooting, in-house system design and build, and Python automation. Founder of Robertech, a custom software engineering studio now in development. I own problems end to end: requirements, architecture, implementation, testing, deployment and handover documentation.

WHAT I DO

- **Full-Stack Web Applications.** Complete web products from concept to deployment — responsive front ends over secure backend APIs.
- **Backend Systems & Database Architecture.** Data modelling, CRUD, auth and REST endpoints in Node.js, Python or PHP, with defensive error handling.
- **Custom Business Software & GUIs.** Desktop management tools, POS systems and internal operational software on object-oriented architectures.
- **Frontend Architecture & UI Redesign.** Slow or dated interfaces rebuilt fast and accessible (WCAG AA) on design-token systems.
- **Linux Systems & DevOps Automation.** Server provisioning, Bash scripts, Terraform, Nginx proxying, load balancing, firewall hardening, CI/CD.
- **Technical Consultation & Code Review.** Stack selection, performance-bottleneck and database-optimisation reviews, engineering guidance.

How I work — 01 Discovery and requirements alignment · 02 Architecture, schemas, API contracts and design tokens · 03 Iterative build in milestones with clean Git history and staging demos · 04 Quality audit, accessibility and responsive testing, deployment and documented handover.

EXPERIENCE

Founder & Lead Engineer · Robertech

2026 – Present

Custom software engineering studio · Nairobi, Kenya · in development · robadev254-robertecch-vmgj.vercel.app

- ▶ Founded a studio building bespoke business systems instead of reselling packaged products, across five service lines: **point-of-sale, adaptive ERP, tailored business software, web platforms and enterprise automation.**
- ▶ Building the platform end to end in **type-safe TypeScript** over a **schema-validated PostgreSQL** database, with an audit trail on critical state transitions and modular, test-covered code.
- ▶ Engineering the POS core for real retail conditions — high-concurrency checkout, offline resilience with a sync queue, multi-branch replication to HQ and fast barcode scanning — and owning positioning, the client-facing site, hosting and Vercel deployment.

IT Support Personnel · Kenya Forest Service (KFS), Headquarters

Aug 2025 – Dec 2025

Industrial attachment · Nairobi, Kenya

- ▶ **Database management** — administered institutional databases: structured queries and reporting extracts, record integrity checks, routine backups and restore verification.
- ▶ **Network installation & troubleshooting** — installed and configured LAN infrastructure across HQ departments (cabling, switch ports, IP addressing, user access); diagnosed connectivity, DNS and hardware faults as first-line support.
- ▶ **System design, build & testing** — gathered requirements from departmental users, designed the data model and workflow, implemented the system and ran functional and acceptance testing before rollout.
- ▶ **Web design** — built web pages and internal interfaces to accessible, responsive HTML/CSS/JavaScript standards.
- ▶ **Automation with Python** — replaced repetitive administrative and reporting routines with Python scripts, cutting hand-keyed data entry and report turnaround.
- ▶ **Authorised web scripting** — wrote sanctioned Python scripts to collect, clean and consolidate data from approved web sources into internal reporting formats.

SELECTED PROJECTS — GITHUB.COM/ROBADEV254

Robertech Studio Platform · live

Service catalogue, an interactive sandbox simulating POS, ERP and event-automation telemetry, hosting pages and a command-centre dashboard.

TypeScript · Next.js · PostgreSQL · Vercel

Engineering Portfolio & Design System · RobadevPortfolio

Seven-page site with no CSS or JS framework: custom design tokens, self-hosted variable typography, mail-draft inquiry composer, WCAG AA keyboard and reduced-motion paths.

HTML5 · Vanilla CSS3 · JavaScript ES6+ · Netlify

Restaurant POS & Operations Manager · USERINTERFACE-JAVA

Desktop application for order taking, billing and discounts, receipt generation and menu inventory. Event-driven UI with input sanitisation; data models separated from the view for database persistence.

Java SE · Swing · OOP / MVC

Marketplace & Debt Management · nexus_marketplace, Debt-Management

A marketplace with listings and transactional flows, plus a debt-tracking system on the CodeIgniter 4 MVC framework with ledger records and repayment schedules.

PHP · CodeIgniter 4 · MySQL

Across 36 public repositories — Sakupharmacy pharmacy front end · CAT-ML machine-learning notebooks · CSharp advanced application programming · Banking (C++) · Java Swing GUI builds (Flags, Fan, Divisible-by-6-or-5) · Turtle graphics.

ALX SOFTWARE ENGINEERING PROGRAMME — REPOSITORY PORTFOLIO

1 · Low-Level C & Systems Programming

Memory management with malloc/free and pointer arithmetic, a from-scratch variadic printf, a working Unix shell (fork/exec, PATH resolution, built-ins, signals), a stack/queue bytecode interpreter, and sorting algorithms benchmarked by big-O complexity.

alx-low_level_programming · printf · simple_shell · monty · sorting_algorithms

3 · System Engineering & DevOps

Bash automation, Linux permissions and process management, SSH key configuration, web-stack debugging, Nginx configuration and redirects, load balancing with health checks, SSL/TLS termination, UFW firewall hardening, monitoring and web-infrastructure design write-ups.

alx-system_engineering-devops

Hospital Management System · hospital

Patient registration, patient records and prescription management against a relational schema with server-side validation.

PHP · MySQL · HTML · CSS

2 · Higher-Level Programming (Python)

Python OOP and inheritance, exceptions, unit testing and documentation, JSON/CSV serialisation, file I/O, MySQL through an ORM, and consuming REST APIs from scripts. Includes an RSA factoring challenge exercising number theory and performance-bounded search.

alx-higher_level_programming · RSA-Factoring-Challenge

4 · Toolchain Foundations

Git and GitHub workflow, Vagrant/Ubuntu environment setup, editor and shell fundamentals — the programme's onboarding track.

alx-zero_day · alx-pre_course · zero_day

TECHNICAL SKILLS

Languages	Python · JavaScript (ES6+) · TypeScript · Java · C · C++ · C# · PHP · SQL · Bash
Frontend	React · Next.js · semantic HTML5 · modern CSS3 (Grid, Flexbox) · design-token systems · responsive layout · WCAG AA accessibility
Backend & Data	Node.js / Express · REST API design · authentication & JWT · CodeIgniter 4 · MySQL / MariaDB · PostgreSQL · relational schema design · query optimisation and indexing · migrations and backups
Infrastructure & DevOps	Terraform (infrastructure as code) · Linux / Ubuntu administration · Bash automation · Nginx reverse proxy and load balancing · SSL/TLS · SSH and UFW hardening · GitHub Actions CI/CD · Vercel and Netlify deployment
Networking & IT Support	LAN installation and cabling · switch and router configuration · IP addressing and subnetting · connectivity and DNS troubleshooting · hardware and software diagnostics · end-user support
Automation & Analysis	Python task automation · authorised web scripting and data extraction · report generation · Jupyter notebooks · machine-learning fundamentals · applied AI tooling and prompt engineering
Practices & Strengths	Git and GitHub · OOP and clean code · MVC and SOLID · unit testing · Postman and cURL · Agile milestone delivery · analytical problem solving · clear written and verbal communication · client-facing requirements gathering
Spoken Languages	English — fluent, written and spoken · Kiswahili — fluent, written and spoken
Interests	Coding challenges on LeetCode and HackerRank · hackathons and collaborative builds · open-source contribution · emerging frameworks and developer tooling · tech meetups, webinars and online communities

EDUCATION & CERTIFICATIONS

BSc Software Development · KCA University, Nairobi	Graduated 2025
Second Class Honours, Upper Division · full-time four-year programme · software engineering lifecycles, database management systems, data structures and algorithms, OOP construction, web systems and network fundamentals	

Kenya Certificate of Secondary Education · Makueni Boys High School	Mean grade B+
Sciences, mathematics and analytical problem solving	

- ALX Africa — Software Engineering Programme. 12-month full-time intensive; final score 80%, Distinction. C and low-level programming, Unix system administration, data structures and algorithms, Python, SQL, API architecture and DevOps deployment.
- ALX Africa — AI Career Essentials. Score 85%, Distinction. Applied AI tooling, prompt engineering, data analysis and modern engineering workflows.
- In progress: AWS Certified Cloud Practitioner · Advanced Python Programming.